

AI Ethics Canvas Analysis on Autonomous Systems

CIA 3 – Component 1

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Subject	Responsible AI
Assessment	CIA 3 – Component 1: AI Ethics Canvas Analysis on Autonomous Systems
Case Study Selected	Kargu-2 Autonomous Drone Incident, Libya (March 2020)

Case Background

In March 2020, during the conflict between Libya's UN-recognized Government of National Accord (GNA) and forces loyal to Gen. Khalifa Haftar, Turkish-made STM Kargu-2 loitering munitions were reportedly deployed. According to the UN Panel of Experts on Libya report (2021), the drones were programmed to attack targets without requiring data connectivity between the operator and the munition — a true “fire, forget and find” capability. Logistics convoys and retreating Haftar-affiliated forces were hunted down and remotely engaged by these systems. Notably, the UN report stops short of confirming that human beings were killed by systems operating fully without human supervision — this ambiguity is itself central to the ethical and legal debate the case has generated.

1. Identification of Ethical Issue

The central ethical issue is delegated lethal decision-making without meaningful human control — often called the “responsibility gap.” The Kargu-2 could reportedly select and engage targets autonomously, removing a human from the loop at the critical moment of the kill decision.

This raises the core question of whether a machine should be permitted to make the final judgment to end a human life, and whether existing International Humanitarian Law (IHL) — which requires distinguishing active combatants from those hors de combat (wounded, surrendering, or incapacitated) — can be reliably applied by an image-classification algorithm operating in real time on a battlefield.

A second, closely related ethical dimension is the loss of moral agency in warfare. Traditional just war theory assumes a human combatant weighs proportionality, necessity, and distinction before using lethal force. When a “fire, forget and find” system is launched, that moral deliberation happens once, in advance, at the software and policy level — not at the moment of the kill itself. This substitutes a one-time programming decision for the continuous, context-sensitive judgment IHL was designed to require.

A third dimension is the transparency and explainability problem: even the system's operators may not be able to reconstruct why a specific target was selected, since the classification logic is embedded in a trained model rather than a transparent rule set. This makes it difficult to conduct after-the-fact ethical or legal review of any single engagement.

Finally, the case raises a slippery-slope/precedent concern: because this may be the first documented instance of an AI system autonomously engaging a human target, it functions as a moral and legal test case that could normalize similar deployments by other states or non-state actors with far less oversight than even existed here.

2. Stakeholder Analysis

- Combatants/targets on the ground (Haftar-affiliated forces) — directly at risk of being killed or misidentified by an algorithm with no final human check, particularly while retreating.

- Turkish and GNA military commanders — authorized deployment of the system and set its rules of engagement, but may not have directly authorized each individual engagement once the drone was airborne.
- STM (drone manufacturer) — designed, built, and marketed the autonomous targeting capability, and made the design choice to enable operation without a live data link to a human operator.
- Libyan civilians in the conflict zone — exposed to indirect risk if fighters retreat toward populated areas, or if the classification system misreads civilian activity as hostile.
- International community / United Nations — responsible for monitoring IHL compliance in the conflict and for developing binding norms around lethal autonomous weapons systems (LAWS); their Panel of Experts report is the primary source documenting the incident.
- Future militaries and the arms industry worldwide — this case functions as a real-world precedent and “proof of concept” that other states and manufacturers may study or replicate.
- Human rights and arms-control organizations (e.g., Campaign to Stop Killer Robots, ICRC) — advocacy stakeholders pushing for treaty-level restrictions, whose credibility and leverage are affected by how this case is resolved.
- Families of the deceased or affected fighters — bear the human cost of the incident and have the greatest interest in accountability and redress, yet the least power to obtain it given the attribution problem.

3. Harm and Risk Assessment

- Direct physical harm: possible unlawful killing of retreating or surrendering combatants who, under IHL, should no longer be legitimate targets once they cease to pose a threat.
- Misclassification risk: image-processing systems may struggle to detect subtler surrender or incapacitation signals — a wounded or unconscious person, or someone unable to physically discard a weapon, may not be reliably recognized as hors de combat, increasing the chance of unlawful killing.
- Erosion of legal accountability: without a clear human decision-maker at the point of engagement, victims or their families have no clear path to seek justice, investigation, or redress through military or international courts.
- Proliferation risk: normalizes autonomous targeting for other states and non-state actors, potentially accelerating an arms race in lethal autonomy with inadequate safety testing or oversight.
- Erosion of international humanitarian norms more broadly, since this precedent could be cited by future actors to justify similar or more permissive autonomous-weapon deployments.
- Psychological and societal harm: normalizing machine-initiated killing may lower the perceived threshold for going to war (“risk-free killing” for the deploying side), potentially making armed conflict more frequent or prolonged.
- Data and training bias risk: if the drone's targeting model was trained on limited or unrepresentative combat imagery, it may perform unpredictably in unfamiliar terrain, lighting, or clothing conditions, increasing the risk of misidentification beyond what was tested.
- Escalation and miscalculation risk: autonomous systems acting faster than human decision cycles could trigger unintended escalation between opposing forces before human commanders are even aware an engagement occurred.

4. Responsibility and Accountability Analysis

- STM (manufacturer): responsible for designing a system capable of autonomous lethal engagement and for the reliability, testing, and documented limitations of its target-classification algorithms; arguably bears product-design responsibility similar to a manufacturer's duty of care.
- Turkish military / GNA-affiliated forces: bear command responsibility for authorizing deployment of a system capable of “fire, forget and find” operation, effectively pre-delegating the kill decision before

the mission began, and for the rules of engagement programmed into the system.

- Field commanders and operators: even without a live data link, whoever set the mission parameters, target profile, and geographic engagement zone made consequential decisions that shaped what the drone would ultimately do — responsibility is distributed, not eliminated, by autonomy.
- State-level accountability: under IHL, states remain responsible for weapons used by their forces, but attributing a specific unlawful killing to a specific decision-maker becomes extremely difficult when no human made the final engagement call — the classic “responsibility gap” in AI ethics, where no single party neatly satisfies the legal standard for culpability.
- International regulators (UN, CCW): bear some responsibility for the continued absence of binding international law specifically governing lethal autonomous weapons, a legal vacuum this case exposed and that existing IHL frameworks (designed for human combatants) were not built to address.
- Shared/distributed responsibility model: rather than assigning blame to one party, this case illustrates why AI ethics increasingly favors a distributed accountability model — spanning designer, deployer, commander, and regulator — since traditional single-agent liability frameworks break down when decision-making is spread across humans and machines at different points in time.

5. Recommendations and Ethical Justification

- Mandate meaningful human control (MHC): require a human to authorize each individual lethal engagement, not just the overall mission or deployment, to preserve legal accountability and ensure a moral agent is present at the point of decision.
- Pre-deployment IHL compliance testing: autonomous weapons should be independently tested for their ability to detect hors de combat status (surrender, incapacitation) before being cleared for battlefield use, with published performance thresholds.
- International binding regulation: pursue a treaty under the UN Convention on Certain Conventional Weapons (CCW) that restricts or bans fully autonomous lethal targeting, closing the current normative gap exposed by this case.
- Auditable “black box” logging: require autonomous weapons to log every engagement decision and the sensor data behind it, similar to a flight data recorder, enabling post-hoc accountability investigations and legal review.
- Geofencing and engagement-window limits: restrict autonomous engagement to narrowly defined times, locations, and target profiles set in advance by human commanders, reducing the scope for unsupervised misclassification.
- Mandatory after-action human review: every autonomous engagement should trigger a compulsory post-strike review by military legal/ethics officers, with findings reported to a national or international oversight body.
- Manufacturer disclosure requirements: companies like STM should be required to disclose the training data, testing conditions, and known failure modes of targeting algorithms to end-user militaries and, where appropriate, to international monitors.

Ethical Justification: These recommendations balance legitimate military utility against the ethical principle that lethal-force decisions carry moral weight that should not be fully delegated to a machine. This is consistent with just war theory’s requirement that a responsible, identifiable human agent must stand behind any use of lethal force, and with IHL’s foundational demand for human judgment in distinguishing combatants from protected persons. They also reflect the AI ethics principle of “meaningful human control” increasingly endorsed by the ICRC and UN CCW discussions, and the broader accountability principle that autonomy in a system’s function should never translate into an accountability vacuum for the humans who build and deploy it.

Conclusion

The Kargu-2 incident is widely regarded as one of the first real-world cases where a lethal autonomous weapons system may have engaged human targets without direct human authorization at the moment of attack. Its significance for AI ethics lies less in confirming a body count and more in exposing how quickly technology can outpace the legal and ethical frameworks meant to

govern it. The case demonstrates that “responsible AI” in high-stakes domains cannot be achieved through good intentions alone — it requires enforceable technical constraints (such as mandatory human authorization and auditable logging), corporate transparency from manufacturers, and binding international governance before, not after, such systems are widely deployed. Until such safeguards exist, similar autonomous systems risk repeating the same responsibility gap on a larger scale.

Source: UN Security Council Panel of Experts on Libya (S/2021/229), March 2021; Lieber Institute, West Point, “The Kargu-2 Autonomous Attack Drone: Legal & Ethical Dimensions.”